AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 2: Differential Calculus

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1 Keywords

- Differentiation, linearity, product rule, chain rule, inverse functions, gradients, Jacobians, matrix derivatives, traces

2 Main Points

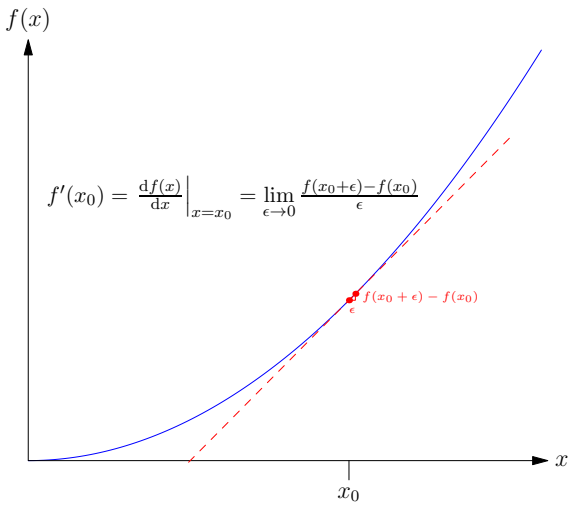
2.1 Why Calculus?

- Differentiation is the fundamental tool of optimisation, and optimisation is how learning machines are trained
- Integration is how we take expectations over continuous distributions
- You met the rules at A-level, but they are easy to forget, and we will use some quite sophisticated tricks. It pays to understand where the tricks come from

2.2 Differentiation from First Principles

- The derivative is the limit of the slope of a chord,

$$f'(x) = \frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{\epsilon\to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}$$



- An equivalent statement, used over and over again in this course, is

$$f(x+\epsilon) = f(x) + \epsilon\, f'(x) + O(\epsilon^2)$$

To find a derivative, expand $f(x+\epsilon)$ and read off the coefficient of ϵ

- **Polynomials.**
$$\frac{\mathrm{d}x^2}{\mathrm{d}x} = \lim_{\epsilon\to 0} \frac{(x^2 + 2\,\epsilon\,x + \epsilon^2) - x^2}{\epsilon} = \lim_{\epsilon\to 0}(2\,x + \epsilon) = 2\,x$$
More generally $(x+\epsilon)^n = x^n + n\,\epsilon\,x^{n-1} + O(\epsilon^2)$, so $\mathrm{d}x^n/\mathrm{d}x = n\,x^{n-1}$

- **Linearity.** Splitting the limit,
$$\frac{\mathrm{d}}{\mathrm{d}x}\left(a\,f(x) + b\,g(x)\right) = a\,f'(x) + b\,g'(x)$$
Differentiation is a linear operation

2.3 Product and Chain Rules

- **Product rule.** Multiplying out the expansions of $f(x+\epsilon)$ and $g(x+\epsilon)$,
$$f(x+\epsilon)\,g(x+\epsilon) = f(x)\,g(x) + \epsilon\left(f'(x)\,g(x) + f(x)\,g'(x)\right) + O(\epsilon^2)$$
so the derivative of $h(x) = f(x)\,g(x)$ is $h'(x) = f'(x)\,g(x) + f(x)\,g'(x)$

- **Chain rule.** For $h(x) = f(g(x))$,
$$h(x+\epsilon) = f\left(g(x) + \epsilon\,g'(x) + O(\epsilon^2)\right) = f(g(x)) + \epsilon\,g'(x)\,f'(g(x)) + O(\epsilon^2)$$
so $h'(x) = g'(x)\,f'(g(x))$

- Together the product and chain rules let us differentiate almost anything

- The chain rule is often easier to remember as
$$\frac{\mathrm{d}f(g(x))}{\mathrm{d}x} = \frac{\mathrm{d}f(g)}{\mathrm{d}g}\,\frac{\mathrm{d}g(x)}{\mathrm{d}x}$$
For example
$$\frac{\mathrm{d}\,\mathrm{e}^{\cos(x^2)}}{\mathrm{d}x} = \frac{\mathrm{d}\,\mathrm{e}^{\cos(x^2)}}{\mathrm{d}\cos(x^2)}\,\frac{\mathrm{d}\cos(x^2)}{\mathrm{d}x^2}\,\frac{\mathrm{d}x^2}{\mathrm{d}x} = \mathrm{e}^{\cos(x^2)}\left(-\sin(x^2)\right)\,2\,x = -2\,x\,\sin(x^2)\,\mathrm{e}^{\cos(x^2)}$$

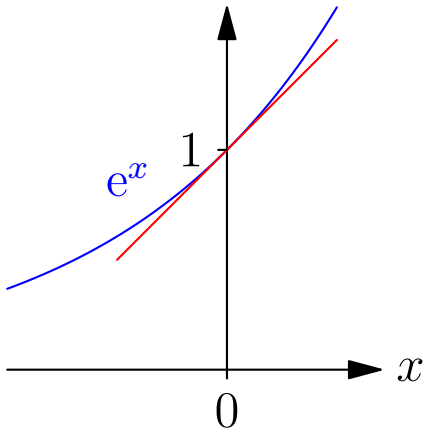
2.4 Inverse Functions

- If $g = f^{-1}$, so $g(f(x)) = x$, differentiating both sides with the chain rule gives $f'(x)\,g'(f(x)) = 1$

- Writing $y = f(x)$,
$$\frac{\mathrm{d}f^{-1}(y)}{\mathrm{d}y} = \frac{1}{f'\left(f^{-1}(y)\right)}$$

2.5 Exponentials and Logarithms

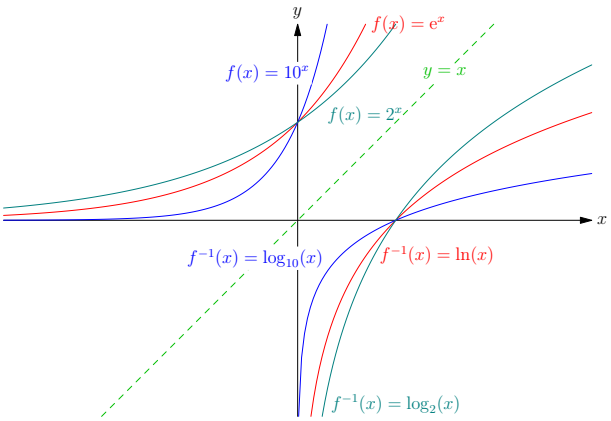
- Since $\mathrm{e}^{\epsilon} = 1 + \epsilon + O(\epsilon^2)$ and $\mathrm{e}^{x+\epsilon} = \mathrm{e}^x\,\mathrm{e}^{\epsilon}$,
$$\mathrm{e}^{x+\epsilon} = \mathrm{e}^x + \epsilon\,\mathrm{e}^x + O(\epsilon^2) \qquad \Longrightarrow \qquad \frac{\mathrm{d}\,\mathrm{e}^x}{\mathrm{d}x} = \mathrm{e}^x$$



- By the chain rule $\mathrm{d}\,e^{c\,x}/\mathrm{d}x = c\,e^{c\,x}$, and more generally $\mathrm{d}\,e^{g(x)}/\mathrm{d}x = g'(x)\,e^{g(x)}$
- For any base, $a^x = e^{x\ln(a)}$, so $\mathrm{d}\,a^x/\mathrm{d}x = \ln(a)\,a^x$
- The natural logarithm is the inverse of the exponential, $\ln(e^x) = x$ and $e^{\ln(y)} = y$. The inverse function rule with $f(x) = e^x$ gives

$$\frac{\mathrm{d}\ln(y)}{\mathrm{d}y} = \frac{1}{e^{\ln(y)}} = \frac{1}{y}$$

- The lecture animation shows e^x and $\ln(x)$ as reflections of one another in the line $y = x$: reflecting a curve swaps rise and run, which is why the slopes of inverse functions are reciprocals



2.6 Derivatives in High Dimensions

- A function $f : \mathbb{R}^n \to \mathbb{R}$ changes at different rates in different directions. The derivative in a unit direction $\boldsymbol{u}$ is the *directional derivative*

$$\partial_{\boldsymbol{u}} f(\boldsymbol{x}) = \lim_{\epsilon \to 0} \frac{f(\boldsymbol{x} + \epsilon\,\boldsymbol{u}) - f(\boldsymbol{x})}{\epsilon}$$

- Taking $\boldsymbol{u} = \boldsymbol{\delta}_i$, the unit vector along the ith axis, gives the partial derivative $\partial f(\boldsymbol{x})/\partial x_i$
- Expanding to first order,

$$f(\boldsymbol{x} + \epsilon\,\boldsymbol{u}) = f(\boldsymbol{x}) + \epsilon\,\boldsymbol{u}^{\mathsf{T}}\boldsymbol{\nabla} f(\boldsymbol{x}) + O(\epsilon^2)$$

where the *gradient* $\boldsymbol{\nabla} f(\boldsymbol{x})$ is the column vector of partial derivatives. This is the start of the high dimensional Taylor expansion

2.7 Computing Gradients

- **Component-wise.** Writing $\boldsymbol{w}^\mathsf{T}\mathbf{M}\,\boldsymbol{w} = \sum_{i,j} w_i\, M_{ij}\, w_j$ and differentiating with respect to w_k picks out the terms with $i = k$ or $j = k$:

$$\frac{\partial}{\partial w_k} \sum_{i,j} w_i\, M_{ij}\, w_j = \sum_j M_{kj}\, w_j + \sum_i w_i\, M_{ik}$$

  so $\boldsymbol{\nabla}\, \boldsymbol{w}^\mathsf{T}\mathbf{M}\,\boldsymbol{w} = \left(\mathbf{M} + \mathbf{M}^\mathsf{T}\right) \boldsymbol{w}$. Tedious, but when you are confused, go back to components

- **By expansion.** A slicker method is to expand $f(\boldsymbol{x} + \epsilon\,\boldsymbol{u})$. For $f(\boldsymbol{x}) = \boldsymbol{x}^\mathsf{T}\mathbf{M}\,\boldsymbol{x} + \boldsymbol{a}^\mathsf{T}\boldsymbol{x}$,

$$\begin{aligned} f(\boldsymbol{x} + \epsilon\,\boldsymbol{u}) &= f(\boldsymbol{x}) + \epsilon \left(\boldsymbol{u}^\mathsf{T}\mathbf{M}\,\boldsymbol{x} + \boldsymbol{x}^\mathsf{T}\mathbf{M}\,\boldsymbol{u} + \boldsymbol{a}^\mathsf{T}\boldsymbol{u}\right) + O(\epsilon^2) \\ &= f(\boldsymbol{x}) + \epsilon\, \boldsymbol{u}^\mathsf{T} \left(\mathbf{M}\,\boldsymbol{x} + \mathbf{M}^\mathsf{T}\boldsymbol{x} + \boldsymbol{a}\right) + O(\epsilon^2) \end{aligned}$$

  using $\boldsymbol{x}^\mathsf{T}\mathbf{M}\,\boldsymbol{u} = \boldsymbol{u}^\mathsf{T}\mathbf{M}^\mathsf{T}\boldsymbol{x}$ and $\boldsymbol{a}^\mathsf{T}\boldsymbol{u} = \boldsymbol{u}^\mathsf{T}\boldsymbol{a}$. Reading off the coefficient, $\boldsymbol{\nabla} f(\boldsymbol{x}) = \mathbf{M}\,\boldsymbol{x} + \mathbf{M}^\mathsf{T}\boldsymbol{x} + \boldsymbol{a}$

2.8 Jacobians and the Chain Rule in High Dimensions

- Consider a composition $h(\boldsymbol{x}) = f(\boldsymbol{g}(\boldsymbol{x}))$ with $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^m$ and $f : \mathbb{R}^m \to \mathbb{R}$. The vector valued $\boldsymbol{g}$ is just m scalar functions $g_1, \ldots, g_m$

- Perturbing each one,

$$\boldsymbol{g}(\boldsymbol{x} + \epsilon\,\boldsymbol{u}) = \boldsymbol{g}(\boldsymbol{x}) + \epsilon\,\mathbf{J}\,\boldsymbol{u} + O(\epsilon^2)$$

  where the *Jacobian* $\mathbf{J}$ is the $m \times n$ matrix with $J_{ij} = \partial g_i(\boldsymbol{x})/\partial x_j$

$$\mathbf{J} = \begin{pmatrix} \frac{\partial g_1(\boldsymbol{x})}{\partial x_1} & \frac{\partial g_1(\boldsymbol{x})}{\partial x_2} & \cdots & \frac{\partial g_1(\boldsymbol{x})}{\partial x_n} \\ \frac{\partial g_2(\boldsymbol{x})}{\partial x_1} & \frac{\partial g_2(\boldsymbol{x})}{\partial x_2} & \cdots & \frac{\partial g_2(\boldsymbol{x})}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_m(\boldsymbol{x})}{\partial x_1} & \frac{\partial g_m(\boldsymbol{x})}{\partial x_2} & \cdots & \frac{\partial g_m X(\boldsymbol{x})}{\partial x_n} \end{pmatrix}$$

- Then, to first order,

$$h(\boldsymbol{x} + \epsilon\,\boldsymbol{u}) = f\left(\boldsymbol{g}(\boldsymbol{x}) + \epsilon\,\mathbf{J}\,\boldsymbol{u}\right) = h(\boldsymbol{x}) + \epsilon\,\boldsymbol{u}^\mathsf{T}\mathbf{J}^\mathsf{T}\boldsymbol{\nabla}_{\boldsymbol{g}} f(\boldsymbol{g}(\boldsymbol{x}))$$

- Comparing with $h(\boldsymbol{x} + \epsilon\,\boldsymbol{u}) = h(\boldsymbol{x}) + \epsilon\,\boldsymbol{u}^\mathsf{T}\boldsymbol{\nabla}_{\boldsymbol{x}} h(\boldsymbol{x})$ gives the chain rule in high dimensions,

$$\boldsymbol{\nabla}_{\boldsymbol{x}} h(\boldsymbol{x}) = \mathbf{J}^\mathsf{T}\, \boldsymbol{\nabla}_{\boldsymbol{g}} f(\boldsymbol{g}(\boldsymbol{x})) \qquad \text{or} \qquad \frac{\partial h(\boldsymbol{x})}{\partial x_i} = \sum_{j=1}^{m} \frac{\partial f(\boldsymbol{g})}{\partial g_j}\, \frac{\partial g_j(\boldsymbol{x})}{\partial x_i}$$

- Backpropagation in a neural network is this rule applied layer by layer: the gradient is passed backwards by multiplying by transposed Jacobians

2.9 Differentiating with Respect to Matrices

- Losses often depend on a matrix of weights, e.g. $L(\mathbf{W}) = (\boldsymbol{a}^\mathsf{T}\mathbf{W}\,\boldsymbol{b} - c)^2$. A minimum is a point $\mathbf{W}^*$ where, for every matrix $\mathbf{U}$, $L(\mathbf{W}^* + \epsilon\,\mathbf{U}) - L(\mathbf{W}^*) = O(\epsilon^2)$

- Define $\partial L(\mathbf{W})/\partial \mathbf{W}$ as the matrix of partial derivatives, with elements $\partial L/\partial W_{ij}$. The same argument as for vectors gives

$$L(\mathbf{W} + \epsilon\,\mathbf{U}) = L(\mathbf{W}) + \epsilon\, \mathrm{tr}\left(\mathbf{U}^\mathsf{T} \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}}\right) + O(\epsilon^2)$$

  where $\mathrm{tr}(\mathbf{U}^\mathsf{T}\mathbf{G}) = \sum_{ij} U_{ij}\, G_{ij}$ is the inner product between matrices

- **Example.** For $L(\mathbf{W}) = (\boldsymbol{a}^\mathsf{T}\mathbf{W}\,\boldsymbol{b} - c)^2$,

$$L(\mathbf{W} + \epsilon\,\mathbf{U}) = L(\mathbf{W}) + 2\,\epsilon \left(\boldsymbol{a}^\mathsf{T}\mathbf{W}\,\boldsymbol{b} - c\right) \boldsymbol{a}^\mathsf{T}\mathbf{U}\,\boldsymbol{b} + O(\epsilon^2)$$

  and $\boldsymbol{a}^\mathsf{T}\mathbf{U}\,\boldsymbol{b} = \sum_{ij} a_i\, U_{ij}\, b_j = \mathrm{tr}(\mathbf{U}^\mathsf{T}\boldsymbol{a}\,\boldsymbol{b}^\mathsf{T})$, so $\partial L/\partial \mathbf{W} = 2\left(\boldsymbol{a}^\mathsf{T}\mathbf{W}\,\boldsymbol{b} - c\right) \boldsymbol{a}\,\boldsymbol{b}^\mathsf{T}$

2.10 Traces

- The trace is the sum of the diagonal elements, $\operatorname{tr}\mathbf{A} = \sum_i A_{ii} = \operatorname{tr}\mathbf{A}^\mathsf{T}$
- It is linear, $\operatorname{tr}(c\,\mathbf{A}) = c\operatorname{tr}\mathbf{A}$ and $\operatorname{tr}(\mathbf{A}+\mathbf{B}) = \operatorname{tr}\mathbf{A} + \operatorname{tr}\mathbf{B}$
- $\operatorname{tr}(\mathbf{A}\,\mathbf{B}) = \sum_{ij} A_{ij}\,B_{ji} = \operatorname{tr}(\mathbf{B}\,\mathbf{A})$, so a trace is unchanged by cycling the matrices in a product:

$$\operatorname{tr}(\mathbf{A}\,\mathbf{B}\,\mathbf{C}\,\mathbf{D}) = \operatorname{tr}(\mathbf{D}\,\mathbf{A}\,\mathbf{B}\,\mathbf{C}) = \operatorname{tr}(\mathbf{C}\,\mathbf{D}\,\mathbf{A}\,\mathbf{B}) = \operatorname{tr}(\mathbf{B}\,\mathbf{C}\,\mathbf{D}\,\mathbf{A})$$

- **Quick matrix differentiation.** Since $\partial_\mathbf{U} f(\mathbf{X}) = \operatorname{tr}\left(\mathbf{U}^\mathsf{T}\,\partial f(\mathbf{X})/\partial\mathbf{X}\right)$, expand, then use the trace rules to move $\mathbf{U}^\mathsf{T}$ to the front. For example

$$\partial_\mathbf{U} \operatorname{tr}(\mathbf{A}\,\mathbf{X}\,\mathbf{B}) = \operatorname{tr}(\mathbf{A}\,\mathbf{U}\,\mathbf{B}) = \operatorname{tr}(\mathbf{B}^\mathsf{T}\mathbf{U}^\mathsf{T}\mathbf{A}^\mathsf{T}) = \operatorname{tr}(\mathbf{U}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{B}^\mathsf{T})$$

  so $\partial \operatorname{tr}(\mathbf{A}\,\mathbf{X}\,\mathbf{B})/\partial\mathbf{X} = \mathbf{A}^\mathsf{T}\mathbf{B}^\mathsf{T}$
- Many more useful results are collected in *The Matrix Cookbook*

3 Exercises

3.1 First principles

- Show from the definition that $\mathrm{d}(1/x)/\mathrm{d}x = -1/x^2$

3.2 Chain rule

- Differentiate x^x (hint: write it as an exponential)

3.3 Inverse functions

- Use the inverse function rule to find $\mathrm{d}\arcsin(y)/\mathrm{d}y$

3.4 The logistic function

- Show that $\sigma(x) = 1/(1+\mathrm{e}^{-x})$ satisfies $\sigma'(x) = \sigma(x)\,(1-\sigma(x))$

3.5 Least squares

- By expanding $f(\boldsymbol{w}+\epsilon\,\boldsymbol{u})$, find the gradient of $f(\boldsymbol{w}) = \|\boldsymbol{y} - \mathbf{X}\,\boldsymbol{w}\|^2$

3.6 A matrix derivative

- Find $\partial L/\partial\mathbf{W}$ for $L(\mathbf{W}) = \|\mathbf{W}\,\boldsymbol{x} - \boldsymbol{y}\|^2$

4 Answers

4.1 First principles

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$$\frac{1}{\epsilon}\left(\frac{1}{x+\epsilon} - \frac{1}{x}\right) = \frac{1}{\epsilon}\,\frac{x-(x+\epsilon)}{x\,(x+\epsilon)} = -\frac{1}{x\,(x+\epsilon)} \to -\frac{1}{x^2}$$

4.2 Chain rule

- $x^x = \mathrm{e}^{x\ln(x)}$, so by the chain and product rules

$$\frac{\mathrm{d}x^x}{\mathrm{d}x} = (\ln(x)+1)\,\mathrm{e}^{x\ln(x)} = (\ln(x)+1)\,x^x$$

4.3 Inverse functions

- With $f = \sin$ and $f' = \cos$,

$$\frac{\mathrm{d}\arcsin(y)}{\mathrm{d}y} = \frac{1}{\cos(\arcsin(y))} = \frac{1}{\sqrt{1-y^2}}$$

using $\cos(\theta) = \sqrt{1-\sin^2(\theta)}$, which is non-negative on the range of $\arcsin$

4.4 The logistic function

- $$\sigma'(x) = \frac{\mathrm{e}^{-x}}{(1+\mathrm{e}^{-x})^2} = \frac{1}{1+\mathrm{e}^{-x}}\,\frac{\mathrm{e}^{-x}}{1+\mathrm{e}^{-x}} = \sigma(x)\,(1-\sigma(x))$$

since $1-\sigma(x) = \mathrm{e}^{-x}/(1+\mathrm{e}^{-x})$

4.5 Least squares

- $f(\boldsymbol{w}+\epsilon\,\boldsymbol{u}) = \|(\boldsymbol{y}-\mathbf{X}\,\boldsymbol{w}) - \epsilon\,\mathbf{X}\,\boldsymbol{u}\|^2 = f(\boldsymbol{w}) - 2\,\epsilon\,\boldsymbol{u}^\mathsf{T}\mathbf{X}^\mathsf{T}(\boldsymbol{y}-\mathbf{X}\,\boldsymbol{w}) + O(\epsilon^2)$
- So $\boldsymbol{\nabla} f(\boldsymbol{w}) = -2\,\mathbf{X}^\mathsf{T}(\boldsymbol{y}-\mathbf{X}\,\boldsymbol{w}) = 2\,\mathbf{X}^\mathsf{T}(\mathbf{X}\,\boldsymbol{w}-\boldsymbol{y})$

4.6 A matrix derivative

- $L(\mathbf{W}+\epsilon\,\mathbf{U}) = \|(\mathbf{W}\,\boldsymbol{x}-\boldsymbol{y}) + \epsilon\,\mathbf{U}\,\boldsymbol{x}\|^2 = L(\mathbf{W}) + 2\,\epsilon\,(\mathbf{W}\,\boldsymbol{x}-\boldsymbol{y})^\mathsf{T}\mathbf{U}\,\boldsymbol{x} + O(\epsilon^2)$
- $(\mathbf{W}\,\boldsymbol{x}-\boldsymbol{y})^\mathsf{T}\mathbf{U}\,\boldsymbol{x} = \mathrm{tr}\left(\mathbf{U}^\mathsf{T}(\mathbf{W}\,\boldsymbol{x}-\boldsymbol{y})\,\boldsymbol{x}^\mathsf{T}\right)$ by the same argument as the example above, so

$$\frac{\partial L}{\partial \mathbf{W}} = 2\,(\mathbf{W}\,\boldsymbol{x}-\boldsymbol{y})\,\boldsymbol{x}^\mathsf{T}$$